

CHANGWEI YAO

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EDUCATION

Carnegie Mellon University (CMU)	Pittsburgh, PA
M.S. in Information Systems	Aug. 2024 - May. 2026 (expected)
• Coursework: Mobile Manipulation, Introduction to Robot Learning (Reinforcement Learning), Learning for 3D Vision	
Hunan University (HNU)	Changsha, China
B.E. in Information Security (Ranking: 1/41)	Sep. 2019 - Jun. 2023

TECHNICAL SKILLS

Programming & Libraries	Python, C/C++, MATLAB, Bash, JavaScript, OpenCV, PCL
Software & Tools	Linux, Docker, Git, CMake, GDB, tmux, SSH, OpenClaw
Physical Robots	KUKA LBR iiwa7, Hello Robot Stretch 3, ARX X5, myCobot 320
Robotics & Simulation	ROS/ROS2, MuJoCo, Isaac Gym, Gazebo, PyBullet, MoveIt

PROFESSIONAL EXPERIENCES

CMU College of Engineering, Research Intern Robot Learning, Reinforcement Learning, VLA	Sep. 2024 - Aug. 2025
CyLab Biometrics Center, Advisor: Marios Savvides	CMU, PA
• Built an automated RL reward design system leveraging Graph-of-Thoughts and vision-language models for long-horizon robotic tasks.	
• Designed a bi-level optimization pipeline integrating LLM-based task decomposition with VLM-based rollout evaluation, eliminating human-in-the-loop reward tuning and improving task success rates by +32.25% on RoboGen while achieving 93.73% success on ManiSkill2.	
• Developed a unified post-training framework for vision-language-action models using context-aware meta co-training to improve cross-task generalization and training efficiency.	
CMU Robotics Institute, Research Intern Interactive Navigation, Semantic Mapping	Sep. 2024 - Mar. 2025
Zhang's Lab, Advisor: Ji Zhang	CMU, PA
• Enabled real-time open-vocabulary semantic mapping by integrating LiDAR-based SLAM with vision-language perception, maintaining long-horizon instance consistency under occlusions and scene changes via a spatio-temporal tracking and update pipeline for dynamic objects.	
• Developed a queryable 4D scene graph representation enabling language-guided navigation and spatio-temporal reasoning in the real world.	
SUSTech Dept. of AIM, Research Staff Imitation Learning, Distillation, Data Augmentation	Jul. 2023 - Jun. 2024
Control & Learning for Robotics and Autonomy Lab, Advisor: Wei Zhang	SUSTech, China
• Deployed an efficient robot control pipeline on the KUKA LBR iiwa7 by applying Consistency Models to Diffusion Policies, compressing multi-step inference into a few sampling steps and achieving 10x+ speedup for real-time, low-latency visuomotor control.	
• Proposed a trajectory-level data augmentation method that minimizes interpolation between demonstration segments, improving synthetic data quality and increasing policy success rates by 4.8% and data generation success by 12.8% on MimicGen benchmarks.	
HNU Dept. of CSEE, Undergrad IoT Security, User Authentication, EM Sensing	Oct. 2021 - Aug. 2022
IoT Sensing and Security Lab, Advisor: Wenqiang Jin	HNU, China
• Built a zero-cost authentication system for heterogeneous IoT devices using human-induced EM coupling and rhythm-based signal patterns, achieving 98.67% accuracy in real-world deployment without requiring additional sensors or specialized hardware.	
• Enabled robust cross-device authentication by extracting user-specific temporal EM signal features, improving resilience to environmental noise and spoofing attempts in heterogeneous IoT settings.	

SELECTED PROJECTS

- HiWE for Imitation Learning** 🤖: Developed a semi-supervised imitation learning data processing pipeline on the ARX X5 with hierarchical waypoint extraction, improving policy success rates by 30–50% on real-world manipulation tasks using only 10% labeled data.
- Deep-RL for Minecraft** 🤖: Built a demonstration-driven Deep Q-Learning pipeline for MineRL tree-chopping, including dataset preprocessing, action-space construction, prioritized replay, and end-to-end training/evaluation in sparse-reward environments.
- 4D Human Reconstruction** 🤖: Built a monocular 4D human reconstruction pipeline combining 3D tracking with generative augmentation to produce temporally consistent human representations, enabling robust reconstruction under occlusions and complex human motion.

SELECTED PUBLICATIONS

► IEEE ICRA 2026	First Author
RE-GoT: A Bi-Level Language Model Framework for Reinforcement Learning.	(paper 📄)
► RSS 2026	Co-Author
SuperMap: A Spatio-Temporal SLAM System for Visual-Language Navigation	submitted - (demo 🎥)
► ECCV 2026	Co-Author
Neural Physical Watermarking via Differentiable Print–Capture Simulation	submitted
► ACM TOSN 2024	Co-Author
EM-Rhythm: An authentication method for heterogeneous IoT devices	(paper 📄) project 🎥